

Sorting Algorithm Visualizer

An **interactive sorting algorithm visualizer** built with **Pygame** · watch algorithms sort data in real-time with color-coded animations.

· Features

- **Visual Rendering** · Color-coded bar charts showing sorting progress
 - **Real-time Animation** · Watch swaps and comparisons as they happen
 - **Configurable** · Adjustable number of elements and window size
 - **Pygame-based** · Smooth rendering with `DrawInformation` class
-

.. Architecture

```
Random Data · Pygame Window · Sorting Algorithm · Real-time Bar Chart
Updates
    ·
    DrawInformation class
    (manages rendering state)
```

· Getting Started

```
cd Sorting-Algorithm-Visualizer
pip install pygame
python tutorial.py
```

· Logic Flow

- 1 **Initialize** · `DrawInformation` class sets up Pygame window and rendering params
- 2 **Generate** · Random data array created
- 3 **Render** · Bars drawn proportional to values with gradient colors
- 4 **Sort** · Algorithm executes with yield/callback for each swap
- 5 **Update** · Pygame redraws bars after each operation
- 6 **Complete** · Final sorted state displayed

Key Class: DrawInformation

Attribute	Purpose
<code>width, height</code>	Window dimensions
<code>lst</code>	Current data array
<code>block_width</code>	Calculated bar width
<code>start_x</code>	Drawing offset
<code>GRADIENTS</code>	Color scheme for bars

· Dependencies

pygame

· License

Educational project · use freely for learning and reference.